

Abhinav Sriharsha Anumanchi

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EDUCATION

San Jose State University

Master of Science in Software Engineering

Jan 2024 – Dec 2025

San Jose, California

- Coursework: Data Mining, Enterprise Software Platforms, System Software, Web UI Design, SW Systems Engineering, Enterprise Distributed Systems, SW Quality Assurance & Testing

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, Java, SQL

AI/ML: LangChain, LangGraph, LLM Fine-Tuning (LoRA, Mistral-7B), LLM Integration (OpenAI, Claude, Gemini), Vector DBs (pgVector, ChromaDB), Prompt Engineering

Backend: FastAPI, Flask, Spring Boot, Node.js, Express.js, REST APIs, GraphQL, Kafka, Microservices, Serverless Architecture

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, BigQuery, Database Optimization

Cloud & DevOps: AWS (Lambda, EC2, S3, Bedrock, Textract), GCP, Docker, Kubernetes, Terraform (Learning), Jenkins, CI/CD, OAuth/JWT

Data & Automation: Apache Airflow, PySpark, Pandas, NumPy

Tools: Git, JIRA, Agile/Scrum, React.js, Next.js, Figma, Postman, Elasticsearch/Kibana

EXPERIENCE

Software Engineer | Altrion.ai

Feb 2026 – Present

Tech Stack: Python, FastAPI, PostgreSQL, React.js, TypeScript, Plaid API, Redis, Docker, AWS

San Jose, CA

- Integrated the full suite of **Plaid API** products (accounts, transactions, investments, liabilities, identity, and crypto) designing a **normalized PostgreSQL schema** from scratch to unify structurally incompatible asset types into a single queryable portfolio layer.
- Owned the full **FastAPI** backend contract end-to-end, wiring all API endpoints to the **React.js** frontend with **Pydantic** validation and structured error handling, ensuring reliable data delivery across all portfolio and spending workflows.
- Designed the company landing page in **Figma** and built it in **React.js** using **Claude Code**, applying **lazy loading** and **asset prefetching** to deliver a production-ready marketing site with fast load times and fully responsive layouts.
- Introduced **Redis** caching for user sessions and Plaid response data, cutting **initial portfolio load latency from 3s to under 500ms**. Containerized the full stack with **Docker** on **AWS EC2** for consistent scaling across environments.

Software Engineer | Aviac Technologies

Jan 2023 – Dec 2023

Tech Stack: Java, Spring Boot, React.js, PostgreSQL, MySQL, Hibernate/JPA, Kafka, Docker, Kubernetes, Jenkins

Hyderabad, India

- Designed and maintained **production-grade RESTful APIs** in **Java Spring Boot** for high-volume enterprise SaaS workflows, applying connection pooling, payload optimization, and caching strategies to reduce **API response latency by 30%**.
- Implemented **Apache Kafka**-based async event-driven architecture across microservices, decoupling producers and consumers to ensure fault-tolerant, ordered message delivery under peak load without blocking synchronous workflows.
- Optimized complex **SQL** and **JPQL** queries, indexes, and stored procedures across **PostgreSQL** and **MySQL**, resolving N+1 query issues and reducing data retrieval times for latency-sensitive SaaS workflows by **40%**.
- Managed **Docker** and **Kubernetes** deployments with **Jenkins CI/CD**, writing unit and integration tests to maintain code coverage standards and enforcing quality gates through structured Agile code reviews.
- Implemented **structured logging and distributed tracing** across microservices, enabling end-to-end request visibility and reducing mean time to incident resolution by **35%** through proactive alerting and log aggregation.

Software Engineer Intern | Top Spot Communication

June 2022 – Dec 2022

Tech Stack: React.js, Node.js, Express.js, PostgreSQL, Python, AWS EC2, Sequelize ORM, OAuth

Hyderabad, India

- Identified high manual overhead in print order processing workflows across **2,000+ daily active users**, built **Python** and **JavaScript** automation scripts to eliminate repetitive data entry, reducing manual intervention by **60%** and cutting data entry errors by **45%**.
- Architected and deployed scalable **RESTful APIs** using **Node.js/Express.js** on **AWS EC2** with load balancing to reliably handle **500+ print orders/day**, applying caching and query optimization to achieve **30%** latency reduction and **99.5%** uptime.
- Resolved slow database performance under peak order concurrency by implementing strategic **PostgreSQL** indexing, query plan analysis, and connection pooling, improving data retrieval by **25%** and enabling **50%** higher concurrency under peak load.

PROJECTS

Learneazy.io | AI Learning Platform

Python, Flask, Next.js, TypeScript, LangChain, pgVector, PostgreSQL, Docker, Kubernetes, AWS, CI/CD

- Built an end-to-end **RAG pipeline** using **LangChain agents** and **PyMuPDF** to ingest and process **500+ page** documents, implementing **agentic tool-calling** to dynamically route queries across knowledge sources, achieving **92%** factual accuracy, reducing **LLM hallucination rate by 30%**.
- Optimized **PostgreSQL** with **pgVector** semantic search and strategic indexing, improving search performance by **4x** and sustaining **200+ concurrent users** with sub-second response times. Implemented **automated alerting**, monitoring reducing incident response time by **45%**.
- Containerized the full platform using **Docker** and **Kubernetes** on **AWS** with **Next.js** and **TypeScript** frontend, deploying automated **CI/CD** pipelines that cut deployment time by **60%** and eliminated manual configuration errors across distributed services.

Text-to-SQL Intelligent Query System | Multi-Agent LLM Platform

Python, LangChain, LangGraph, Mistral-7B, LoRA, Google BigQuery, PostgreSQL, Flask, Docker

- Fine-tuned **Mistral-7B** with **LoRA** adapters on domain-specific SQL datasets, achieving **80%** exact-match accuracy on complex enterprise queries and reducing natural language to SQL conversion time by **92%** for non-technical users.
- Designed a **multi-agent orchestration system** using **LangGraph** with **ReAct**-pattern agents for query validation, and schema reasoning, applying **chain-of-thought prompt engineering** to reduce schema mismatches by **60%** and improve query performance by **40%**.
- Deployed production system on **Docker** with **Google BigQuery** and **PostgreSQL** integration, structured logging, automated failover, and a **Flask** REST API layer with rate limiting and auth, achieving **sub-3-second** response times and **99.5%** uptime.